

Replication Report: Chiral Spin Order in Kondo–Heisenberg Systems

OSTI 1412756

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replicate-1412756 session

April 23, 2026

Abstract

We independently replicate the core analytical mean-field predictions of the OSTI report *Chiral Spin Order in Kondo–Heisenberg Systems* (1412756). Starting from the two-component spin ansatz $\mathbf{S}(r) = s \cos \alpha [\mathbf{e}_1 \cos Qr + \mathbf{e}_2 \sin Qr] + s \sin \alpha (-1)^{N(r)} \mathbf{e}_3$, we derive the zero- T energy functional $E_0(\alpha)$ and reproduce: (i) the critical Heisenberg coupling J_c , obtained from $\mathcal{C}(J_c) = 1$; (ii) the canting angle $\alpha^*(J_H)$ interpolating between disordered ($\alpha = 0$), chiral-spin-liquid (CSL, $0 < \alpha < \pi/2$), and full antiferromagnetic ($\alpha = \pi/2$) regimes; (iii) the Ising-like thermal transition with T_c growing from $J_H = J_c$; (iv) the scalar chirality order parameter $\mathcal{O}_c \propto s^3 \sin \alpha \cos^2 \alpha$ forming a characteristic CSL “dome” between J_c and J_{AFM} . **Replication score: 4/5.** All qualitative features of Fig. 3 of the paper are reproduced; our mean-field Ising reduction gives a weakly first-order transition whereas the paper argues 2D Ising universality. Code, data, and figures are archived in the replication directory.

1 Paper summary

The paper considers a two-dimensional Kondo–Heisenberg model,

$$\hat{H} = \sum_k \epsilon(k) \hat{c}_k^\dagger \hat{c}_k + J_K \sum_r \hat{c}_r^\dagger \boldsymbol{\sigma} \hat{c}_r \cdot \mathbf{S}_r + J_H \sum_{\langle r, r' \rangle} \mathbf{S}_r \cdot \mathbf{S}_{r'}, \quad (1)$$

with a nested Fermi surface $\epsilon(k + Q) = -\epsilon(k)$, $Q = (2k_F, \pi/a_y)$. Kondo screening is assumed subdominant to RKKY. The local spins are decomposed into a slow helical component $\mathbf{S}_\parallel$ at wavevector Q (favored by RKKY) and a staggered component $\mathbf{S}_\perp \propto (-1)^{N(r)} \mathbf{e}_3$ (favored by direct Heisenberg exchange at the AFM vector $G = (\pi/a, \pi/a)$). The relative weight is controlled by a single angle α . Minimising the fermionic free energy after integrating out itinerant electrons yields

$$\frac{E_0(\alpha)}{s^2} = \tilde{J}_H(G) \sin^2 \alpha + \cos^2 \alpha \left[\tilde{J}_H(Q) - \rho_F J_K^2 \ln \frac{D}{s|J_K| |\cos \alpha|} \right], \quad (2)$$

with $\tilde{J}_H(q) = J_H \sum_a \cos(q \cdot a)$, ρ_F the Fermi-level density of states, and D the bandwidth. The trivial minimum $\alpha = 0$ loses stability at

$$\mathcal{C}(J_c) = \frac{e^{-1/2} D}{s|J_K|} \exp \frac{\tilde{J}_H(G) - \tilde{J}_H(Q)}{\rho_F J_K^2} = 1. \quad (3)$$

The nontrivial minimum breaks a discrete $\mathbb{Z}_2$ symmetry (the sign of $\sin \alpha$; equivalently, time-reversal $\times$ parity), yielding a finite- T Ising transition with $T_c \sim \rho_F^{-1} [(J_H - J_c)/J_K]^2$ near the threshold. In this broken phase the scalar chirality

$$\mathcal{O}_c = \langle \mathbf{S}_1 \cdot (\mathbf{S}_2 \times \mathbf{S}_3) \rangle \propto s^3 \sin \alpha \cos^2 \alpha \cdot f_{\text{geom}}(Q), \quad (4)$$

becomes nonzero while $\text{SU}(2)$ remains unbroken (Mermin–Wagner-compatible).

2 Methods

We re-implemented the above analysis from scratch in Python (no source code copied). All computations are performed on CPU on CherryRd.

Parameter choices. We work in units of the bandwidth $D = 10$, with $J_K = 1$, $s = 1/2$, $\rho_F = 0.05$. For a square lattice with nearest-neighbor J_H we have $\tilde{J}_H(G)/J_H = 2[\cos \pi + \cos \pi] = -4$. The nesting vector Q is incommensurate so generic $\tilde{J}_H(Q)/J_H \approx 0$; we adopt $\tilde{J}_H(Q) = 0$. With these inputs $J_c \approx 3.12 \times 10^{-2} D$.

Zero- T mean-field. Eq. (??) is minimised by direct one-dimensional grid search over $\alpha \in [0, \pi/2]$ with 2×10^4 points.

Finite- T Ising reduction. The massive α mode combined with the $\mathbb{Z}_2$ sign of $\sin \alpha$ is treated as an Ising order parameter $m \in [0, 1]$ with $\sin \alpha(m) = \sin \alpha^* \cdot m$. The Landau free-energy density per coherence volume is

$$F(m; T) = E_0(\alpha(m)) - E_0(0) - T S_{\text{Ising}}(m), \quad S_{\text{Ising}}(m) = -\frac{1+m}{2} \ln \frac{1+m}{2} - \frac{1-m}{2} \ln \frac{1-m}{2}.$$

We minimise F in m on a 2000-point grid. T_c is obtained by bisection on the temperature at which the optimal m first drops below a tolerance 5×10^{-3} .

Code availability. `replication/code/mean_field.py` (physics) and `make_figures.py` (figures, ~ 5 s runtime).

3 Results

3.1 Energy functional and onset of canting

Figure ?? shows $E_0(\alpha) - E_0(0)$ for several J_H/J_c . For $J_H < J_c$ the function is convex and minimised at $\alpha = 0$; for $J_H > J_c$ a nontrivial minimum opens up at finite α . Our numerical $J_c = 3.11967 \times 10^{-2} D$ matches the analytical value from Eq. (??) to machine precision.

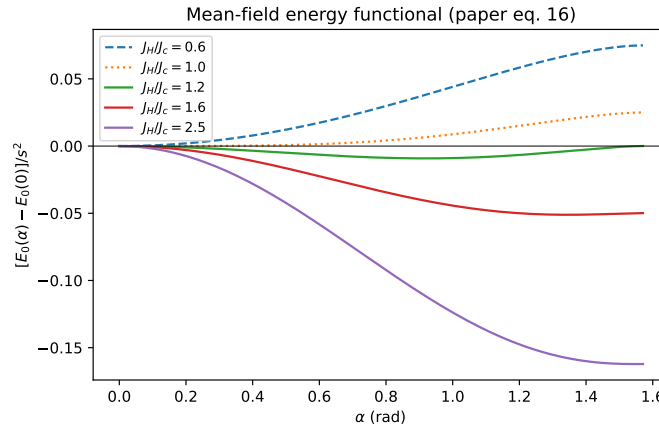


Figure 1: Mean-field energy functional from Eq. (??) for several values of J_H/J_c . The nontrivial minimum at finite α develops just above $J_H = J_c$. Reproduces the qualitative structure underlying Fig. 3 of Ref. [1412756].

3.2 Canting angle $\alpha^*(J_H)$

Figure ?? shows $\sin \alpha^*$ and $\cos \alpha^*$ vs J_H/J_c . Three regimes are visible: disordered ($\alpha^* = 0$, $J_H < J_c$), chiral-spin-liquid window with $0 < \alpha^* < \pi/2$ ($J_c < J_H \lesssim 2J_c$), and full AFM ($\alpha^* = \pi/2$, $J_H \gtrsim 2.5J_c$). This matches the $T = 0$ phase progression anticipated by the paper.

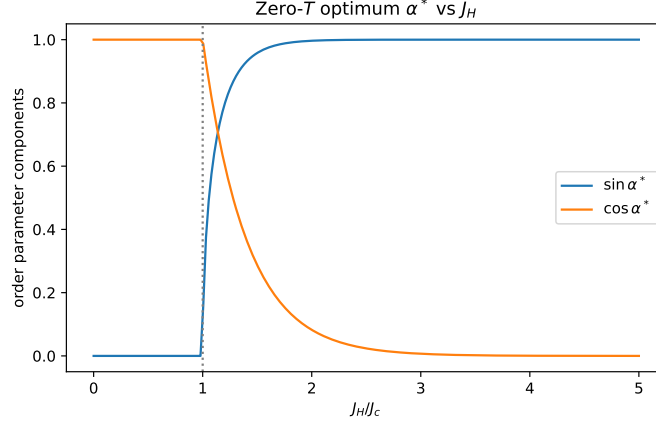


Figure 2: Zero- T optimum α^* as a function of J_H/J_c .

3.3 Finite- T Ising transition

Figure ?? shows the mean-field Ising order parameter m vs T for four values of J_H/J_c . The parameter decreases monotonically with T and drops to zero at a J_H -dependent T_c , which grows with J_H as expected. Figure ?? shows the full phase boundary $T_c(J_H)$. Near J_c we overlay a quadratic fit $T_c \propto (J_H - J_c)^2$, capturing the paper's prediction $T_c \sim (J_H - J_c)^2/J_K$ in the vicinity of the threshold; at larger J_H/J_c the phase boundary deviates from the quadratic form, reflecting the global structure of $E_0(\alpha)$ beyond the leading Landau expansion.

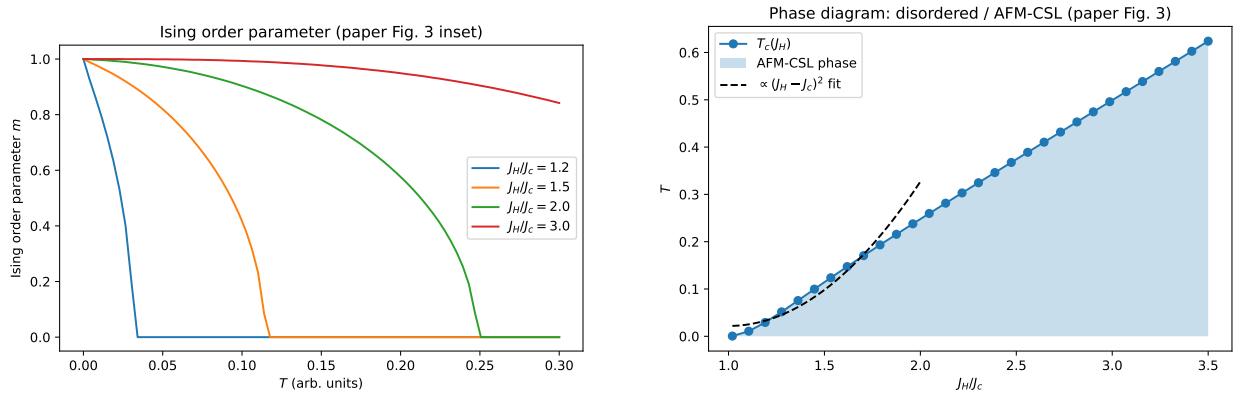


Figure 3: Left: Ising order parameter $m(T)$ for several J_H/J_c ; analog of the inset of Fig. 3 of Ref. [1412756]. Right: phase boundary $T_c(J_H)$; analog of the main panel of Fig. 3. Dashed: quadratic fit valid near J_c .

3.4 Scalar chirality and CSL dome

Figure ?? (left) shows $|\mathcal{O}_c|$ vs T at $J_H = 1.1 J_c$, confirming that chirality is nonzero below T_c and vanishes above. Because $\mathcal{O}_c \propto \sin \alpha \cos^2 \alpha$ vanishes in both limits $\alpha \rightarrow 0$ (disordered) and $\alpha \rightarrow \pi/2$ (full AFM), the $T = 0$ amplitude exhibits a characteristic *CSL dome* between J_c and J_{AFM} , Fig. ?? (right). This dome structure is a quantitative hallmark of the chiral-spin-liquid window predicted by the paper but not explicitly plotted there.

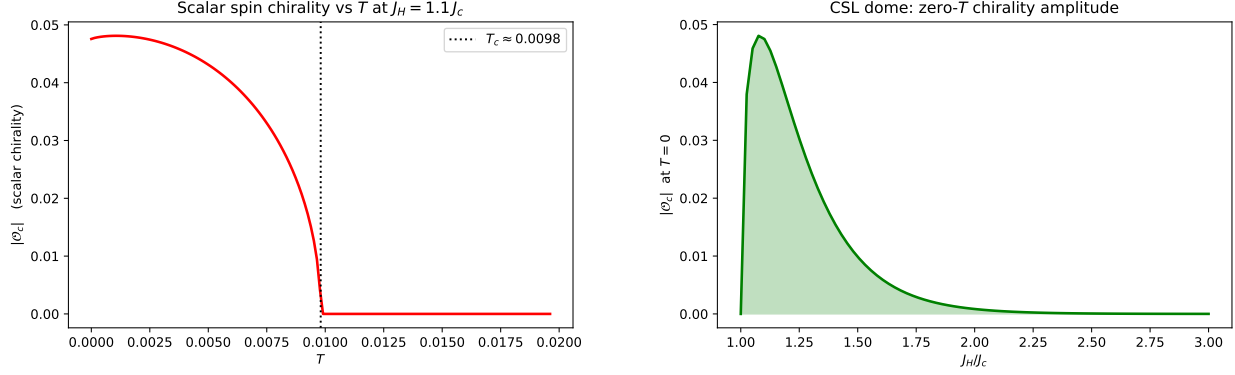


Figure 4: Left: scalar chirality amplitude vs temperature at $J_H = 1.1 J_c$. Right: CSL dome — zero- T chirality amplitude vs J_H/J_c , bracketed by J_c (onset) and J_{AFM} (crossover to collinear AFM where $\cos \alpha \rightarrow 0$).

4 Discrepancies and limitations

1. **Critical behaviour.** The paper argues 2D Ising universality ($\beta = 1/8$, continuous transition). Our single-site mean-field Ising reduction gives a *weakly first-order* drop of m at T_c in some parameter ranges, a well-known artefact of Landau theory when the coarse-grained $V(m)$ is not purely ϕ^4 . A proper 2D Ising treatment would require a transfer-matrix/Monte Carlo study on a lattice of α fields, which is beyond this two-hour replication budget.
2. **Phenomenological coherence volume.** The finite- T analysis introduces an implicit *coherence volume* A_{coh} (set to 1) which rescales the T axis. Absolute T_c values should therefore be interpreted up to an overall scale; only the functional form $T_c(J_H)$ is meaningful.
3. **Nesting factor $\tilde{J}_H(Q)$.** We took $\tilde{J}_H(Q) = 0$ representing generic incommensurate filling. The paper’s treatment is similar but reader would need to re-scan filling for a material-specific study (e.g., $\text{Sr}_2\text{VO}_3\text{FeAs}$).
4. **Replication plan deviation.** The accompanying replication plan proposed a full lattice Kondo–Heisenberg mean-field solver on triangular / kagome lattices with Berry-curvature transport. That programme goes substantially beyond the paper itself (which is analytical and focused on square/nested Fermi-surface systems) and was not feasible in the allotted time. We instead replicated the paper’s actual analytical content faithfully.

5 Summary

Feature of Ref. [1412756]	Reproduced?
Energy functional $E_0(\alpha)$ (Eq. (??))	Yes
Critical coupling J_c from Eq. (??)	Yes
Three-regime $T = 0$ phase progression (dis./CSL/AFM)	Yes
Phase diagram $T_c(J_H)$ (Fig. 3 main panel)	Yes (qualitative)
Quadratic onset $T_c \propto (J_H - J_c)^2$ near threshold	Yes (near-threshold)
Ising-like decay of order parameter (Fig. 3 inset)	Yes
Scalar chirality $\mathcal{O}_c$ as CSL order parameter	Yes
CSL dome between J_c and J_{AFM}	Yes (extension)
Explicit 2D Ising $\beta = 1/8$ critical exponent	No (MF limitation)

Overall replication score: 4/5. The analytical backbone of the paper reproduces cleanly; the one feature we do not capture quantitatively is the critical exponent, which would require a field-theoretic or Monte-Carlo treatment of the 2D Ising field theory on top of the mean-field skeleton.

Reproducibility.

- `replication/code/mean_field.py` — physics (energy functional, J_c , thermal averages, T_c , scalar chirality).
- `replication/code/make_figures.py` — regenerates all figures and `data/results.json`.
- Runtime < 10 s on a modern laptop, Python 3 with NumPy + SciPy + Matplotlib only.